

Simulating biomarker and RNA-Seq data

(Part 1 )

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Overview

This lecture will be in 5 parts:

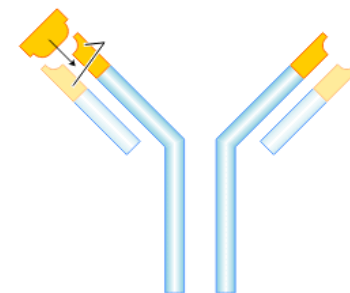
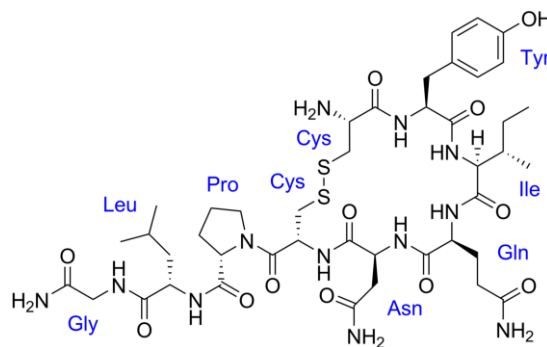
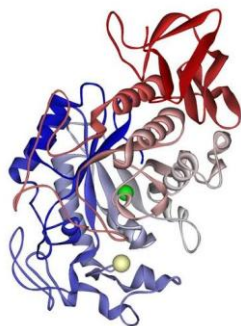
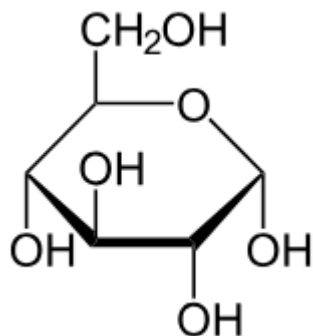
- 1. Definition*
- 2. Why simulate Biomarkers?*
- 3. Biomarker data characteristics*
- 4. Simulation framework*
- 5. Results and interpretation*

1. Definition

What is a biomarker?



*“A **biological molecule** found in **blood, other body fluids, or tissues** that is a sign of a normal or abnormal process, or of a condition or disease. A biomarker may be used to see **how well the body responds to a treatment for a disease or condition**. Also called molecular marker and signature molecule.”*



2. Why simulate Biomarkers?

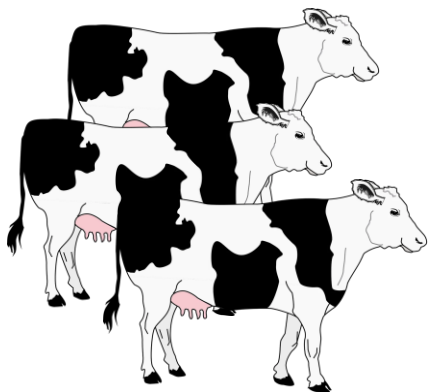
2. *Why simulate Biomarkers?*

- **Prediction of scenarios** 
 - Estimate expected biomarker concentrations under different biological, environmental, or experimental conditions.
 - Explore how biomarkers may respond to treatment, disease, nutrition, stress, or other exposures.
- **Anticipation of biological consequences** 
 - Predict how selection or management decisions may affect biomarker profiles.
 - Explore correlated responses when biomarkers are genetically or biologically associated with other traits.
- **Method validation** 
 - Test statistical models and analytical pipelines under known conditions.
- **Evaluation of experimental designs** 
 - Assess the impact of sample size, group structure, repeated measurements, or variability.

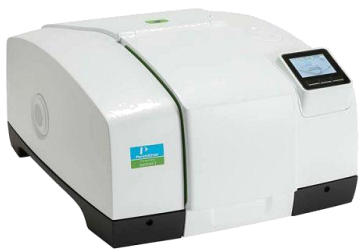
3. Biomarker data characteristics

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- Biomarkers for monitoring physiological imbalance in dairy cows



234 Holstein cows



FT-MIR

Aim

- Evaluate whether easily collected **milk biomarkers** could identify cows in physiological imbalance → risk of metabolic or infectious disease.
- **Compare milk-based measurements** with information normally obtained from more invasive **blood biomarkers**.

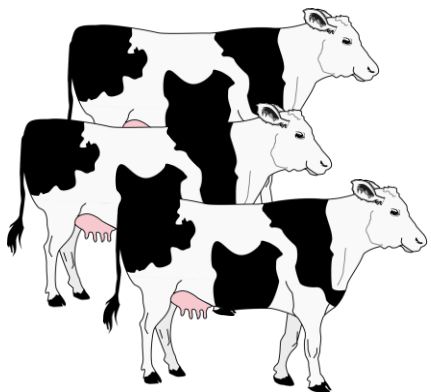
Y = Physiological status characterized using blood glucose, BHB, NEFA and IGF-1, together with energy balance and a physiological imbalance index.

Methods

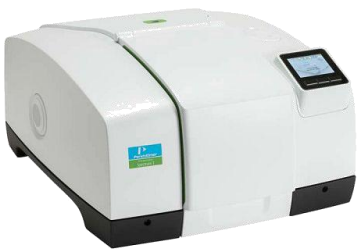
- Compared three groups of milk biomarkers:
 - 8 milk metabolites and enzymes
 - **FT-MIR milk spectra**
 - **19 IgG N-glycans**
- Random Forest models with leave-one-cow-out cross-validation were used for prediction.

3. Biomarker data characteristics

- Biomarkers for monitoring physiological imbalance in dairy cows



234 Holstein cows



FT-MIR

Main findings

- **Milk metabolites/enzymes** and FT-MIR:
 - provided the **best and broadly similar information** about physiological status.
 - predicted the physiological imbalance index with R_{cv}^2 up to **0.40**.
- FT-MIR predicted energy balance somewhat better than the milk metabolites/enzymes.
- Prediction accuracy was **not yet sufficient for reliable identification of individual risk cows**, but milk biomarkers showed potential for non-invasive routine monitoring.

3. Biomarker data characteristics

- Biomarkers as proxies for difficult-to-measure traits

Aim

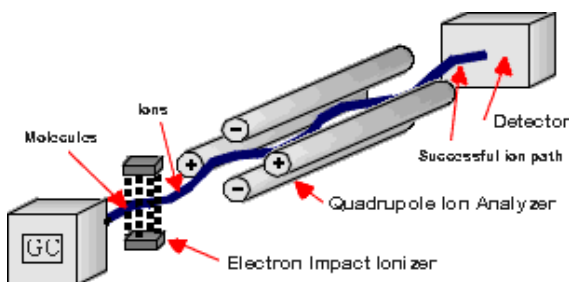
- Determine whether **milk metabolites could predict feed efficiency**, a phenotype that is expensive and difficult to measure routinely.
- Identify a reduced set of metabolomic features capable of predicting **RFI and FCR**.



~40 Assaf ewes

Methods

- Feed efficiency quantified using:
 - Residual feed intake (**RFI**)
 - Feed conversion ratio (**FCR**)
- Untargeted LC-MS generated approximately **3,700 milk metabolomic features**.
- PLS-DA was used for feature selection:
 - **41 features for RFI**
 - **26 features for FCR**



LC-MS

3. Biomarker data characteristics

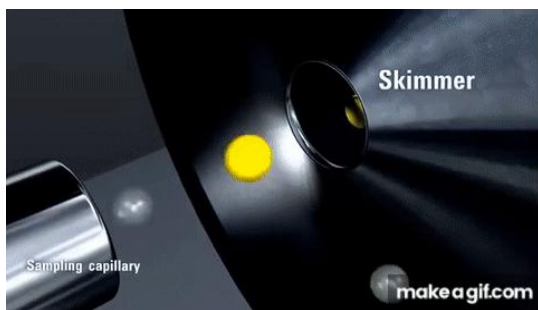
- Biomarkers as proxies for difficult-to-measure traits

Main findings

- Metabolomic profiles clearly **discriminated high-, medium- and low-efficiency** groups, with $AUC > 0.95$.
- **FCR** was considerably easier to predict than RFI.
 - For FCR, the reduced **26-feature signature** reached mean $R^2 \approx 0.62$ with Orthogonal-PLS.
- Candidate biomarkers were linked to **glucose, galactose, fructose, amino-acid, sphingolipid, insulin and thyroid-hormone metabolism**.



~40 Assaf ewes



LC-MS

3. Biomarker data characteristics

Biomarker distributions may be:

- **Binary/categorical** (i.e., mutation present/absent, threshold-defined status).
- **Continuous** (i.e., concentrations, expression levels...)
 - Continuous biomarkers may be **approximately normally distributed**. $Y_i \sim N(\mu, \sigma^2)$
 - However:
 - Their distribution can be affected by biological variation, measurement limits, **detection thresholds**, or the compositional nature of the data.
 - Biomarkers present at very **low concentrations may be close to the limit of detection**, potentially generating censored values or groups of detectable and non-detectable measurements.
 - Relationships between biomarkers and the phenotype **may not be linear**, and neither may the relationships among biomarkers themselves.
 - Many biological concentrations are **positive and right-skewed**, so log transformation is common in biomarker analysis because strongly skewed biomarkers may become approximately Gaussian on the log scale.

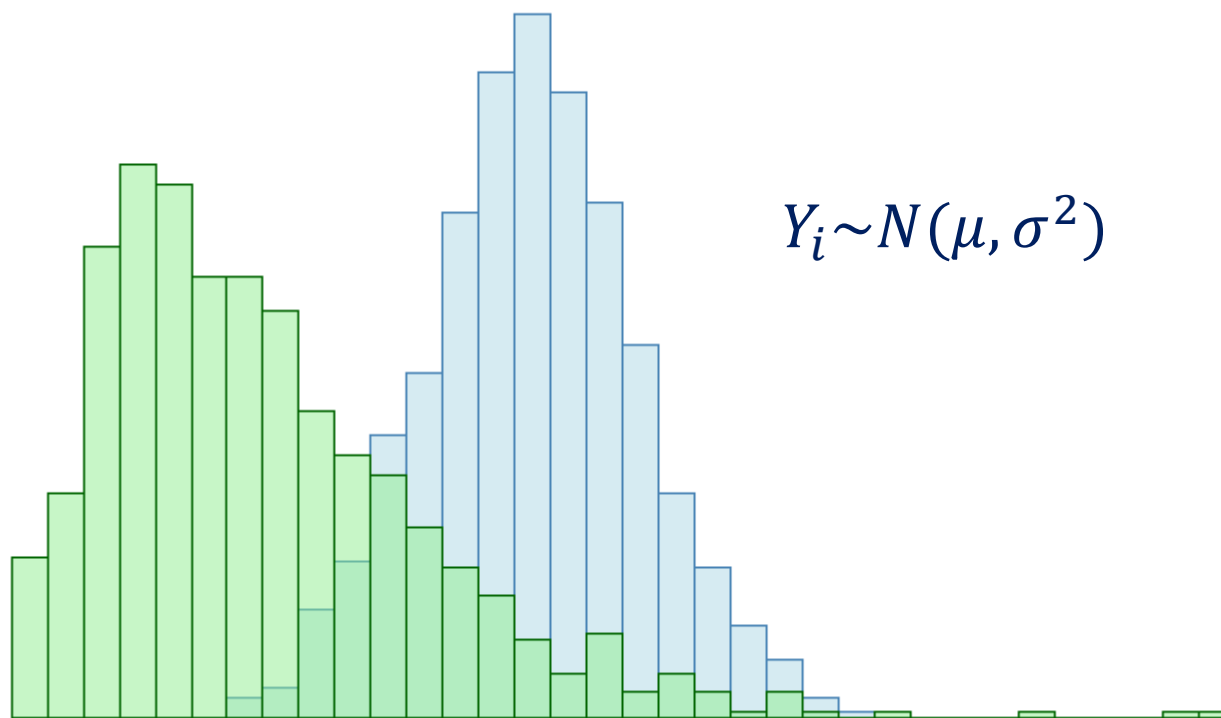
$$\log(Y_i) \sim N(\mu, \sigma^2)$$

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$$\log(Y_i) \sim N(\mu, \sigma^2)$$

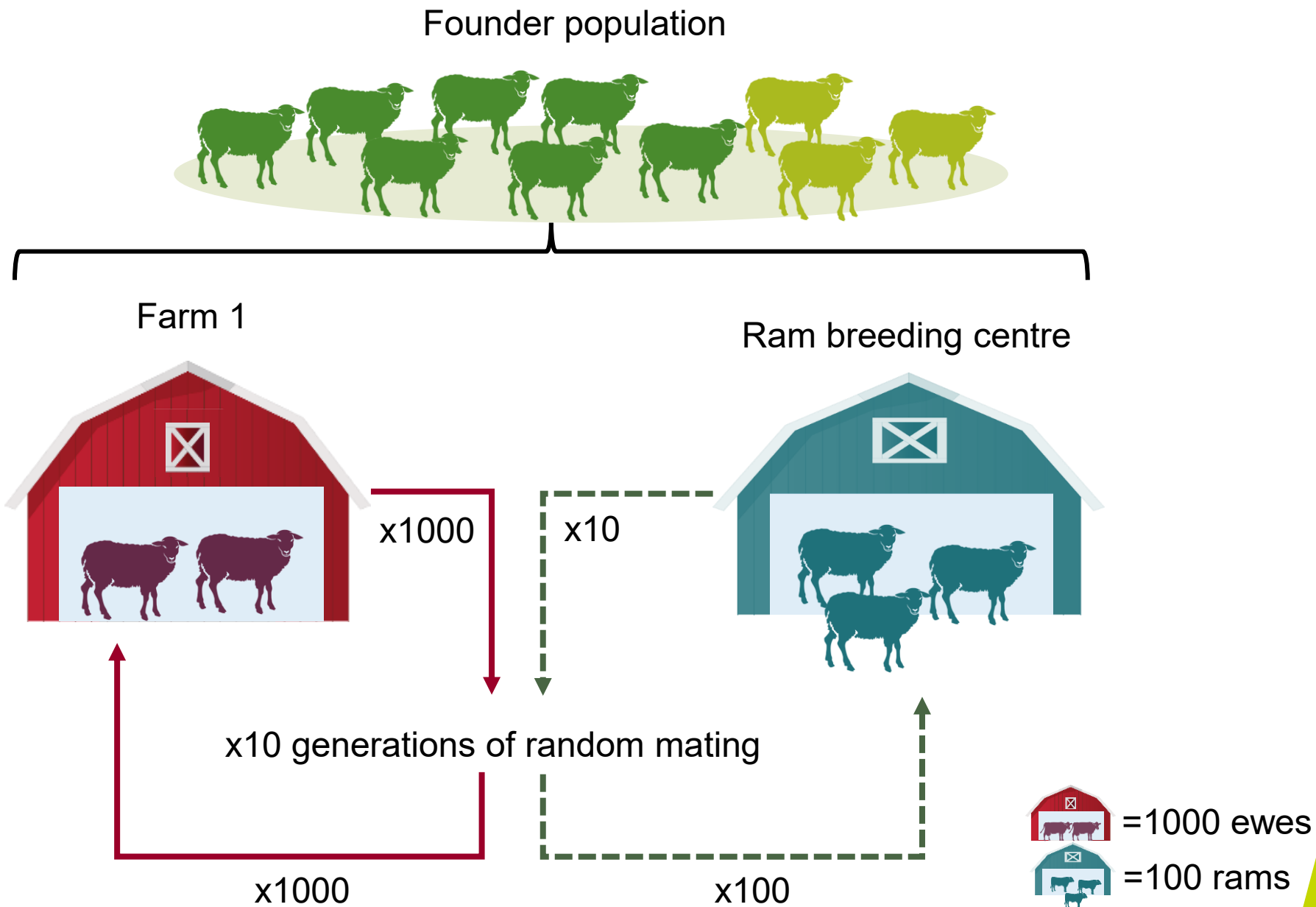


**positive and right-skewed*

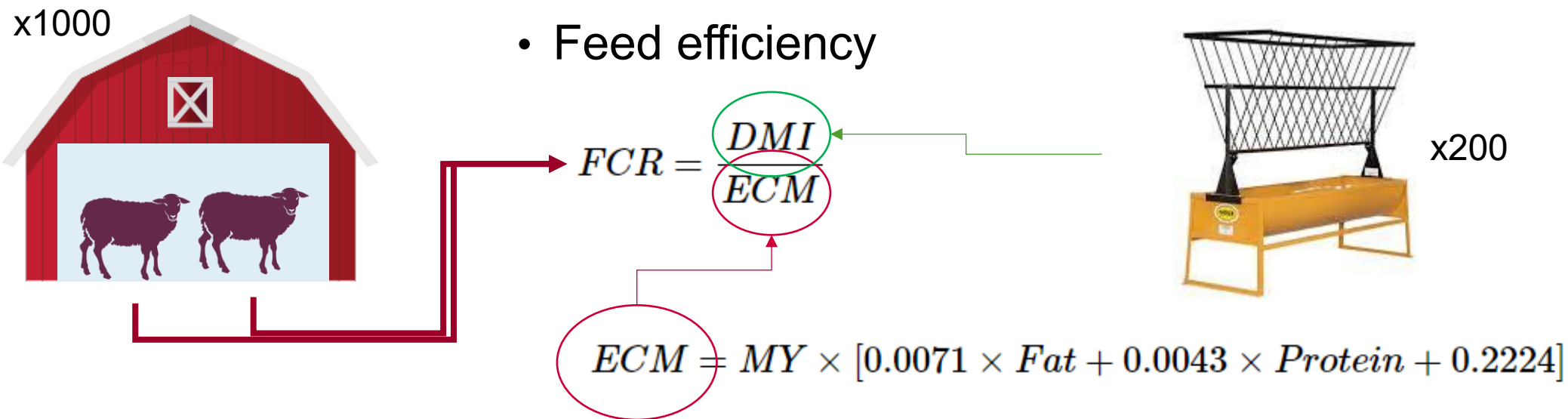
(Grund et al., 2010)

4. Simulation framework

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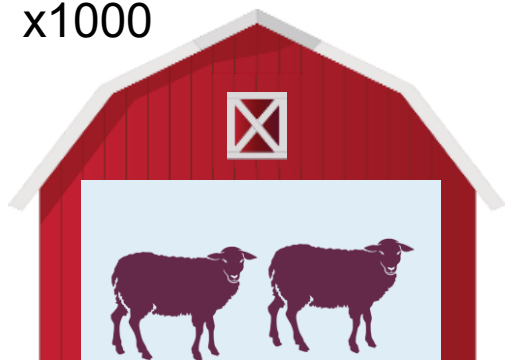
4. Simulation framework



Number of individuals = 50
Control vs treatment groups = 2
Number of biomarkers or genes

4. Simulation framework

x1000

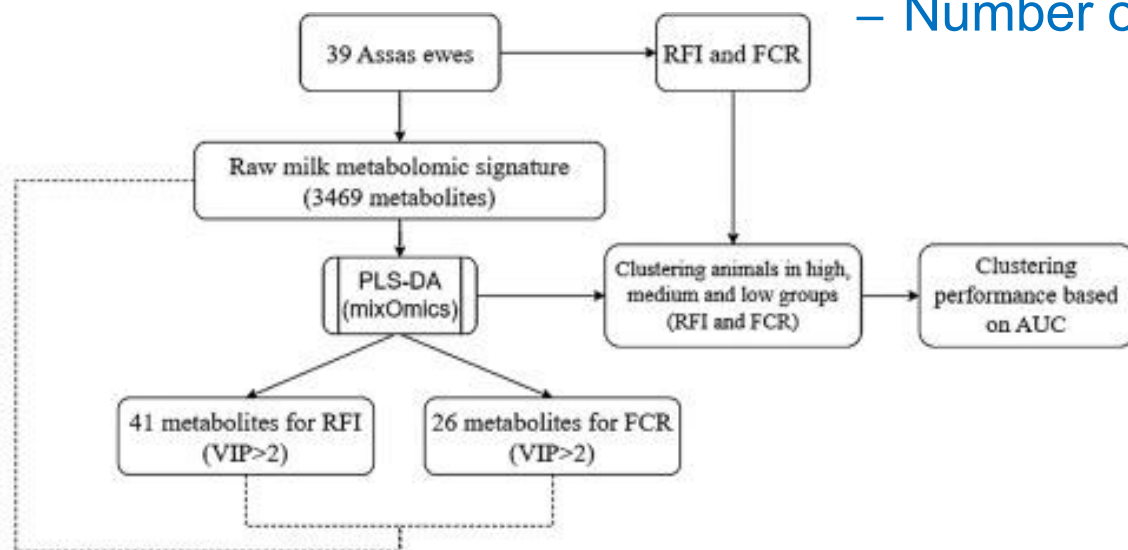


- Feed efficiency in sheep

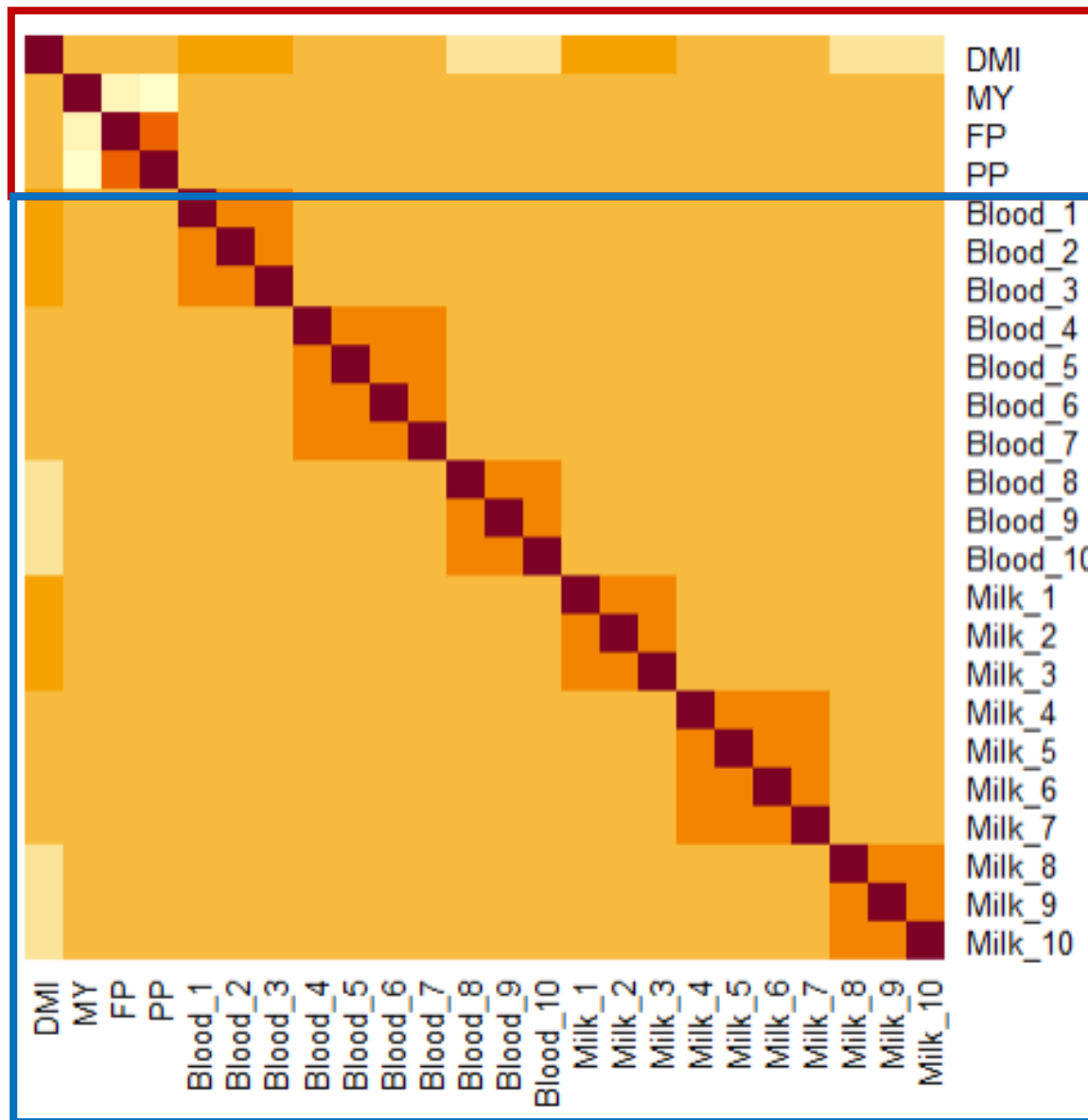
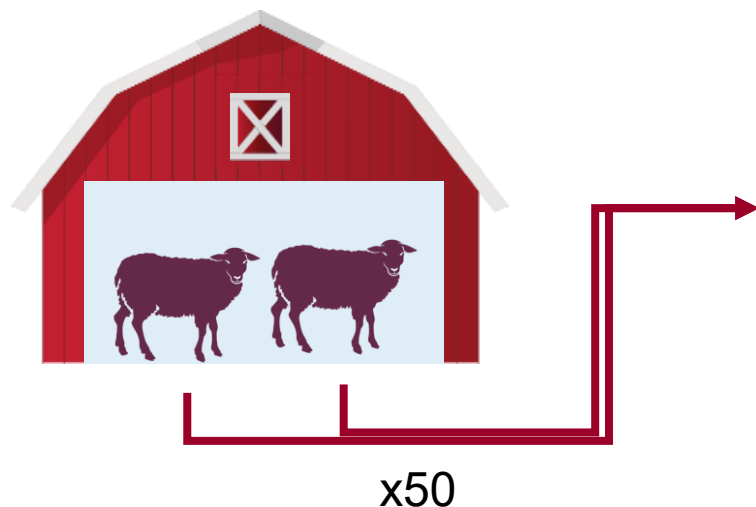
$$FCR = \frac{DMI}{ECM}$$

- Biomarkers

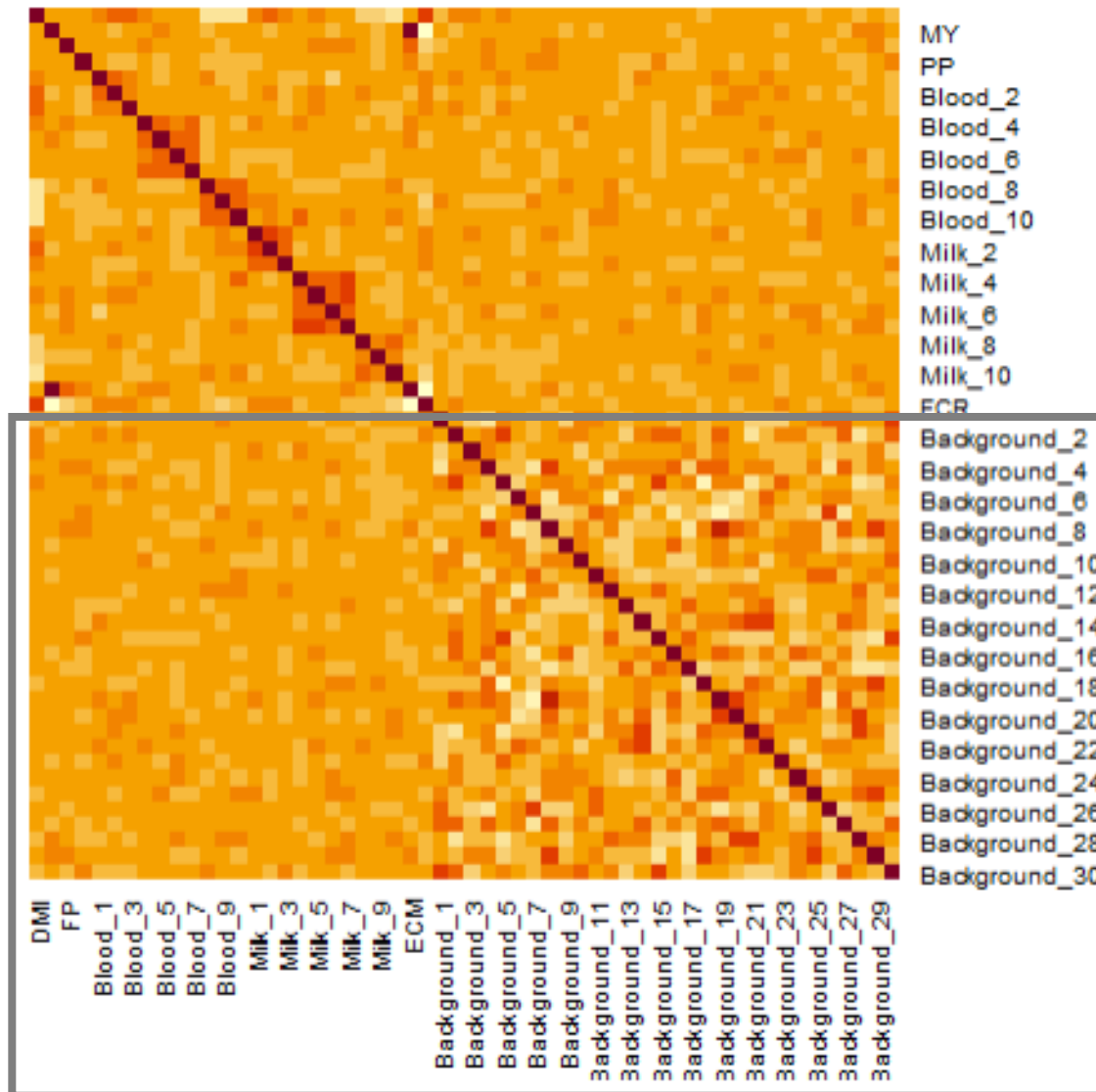
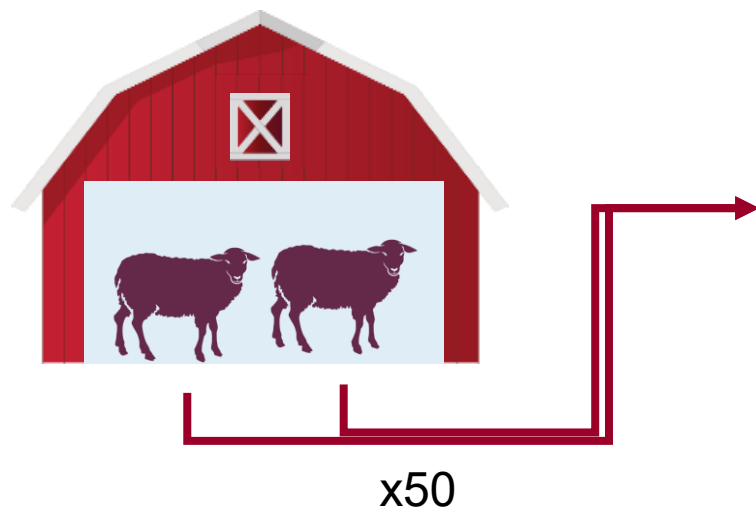
- Number of individuals = 50
- Control vs treatment groups = 2 (HIGH and LOW – FE)
- Number of biomarkers = 50



4. Simulation framework



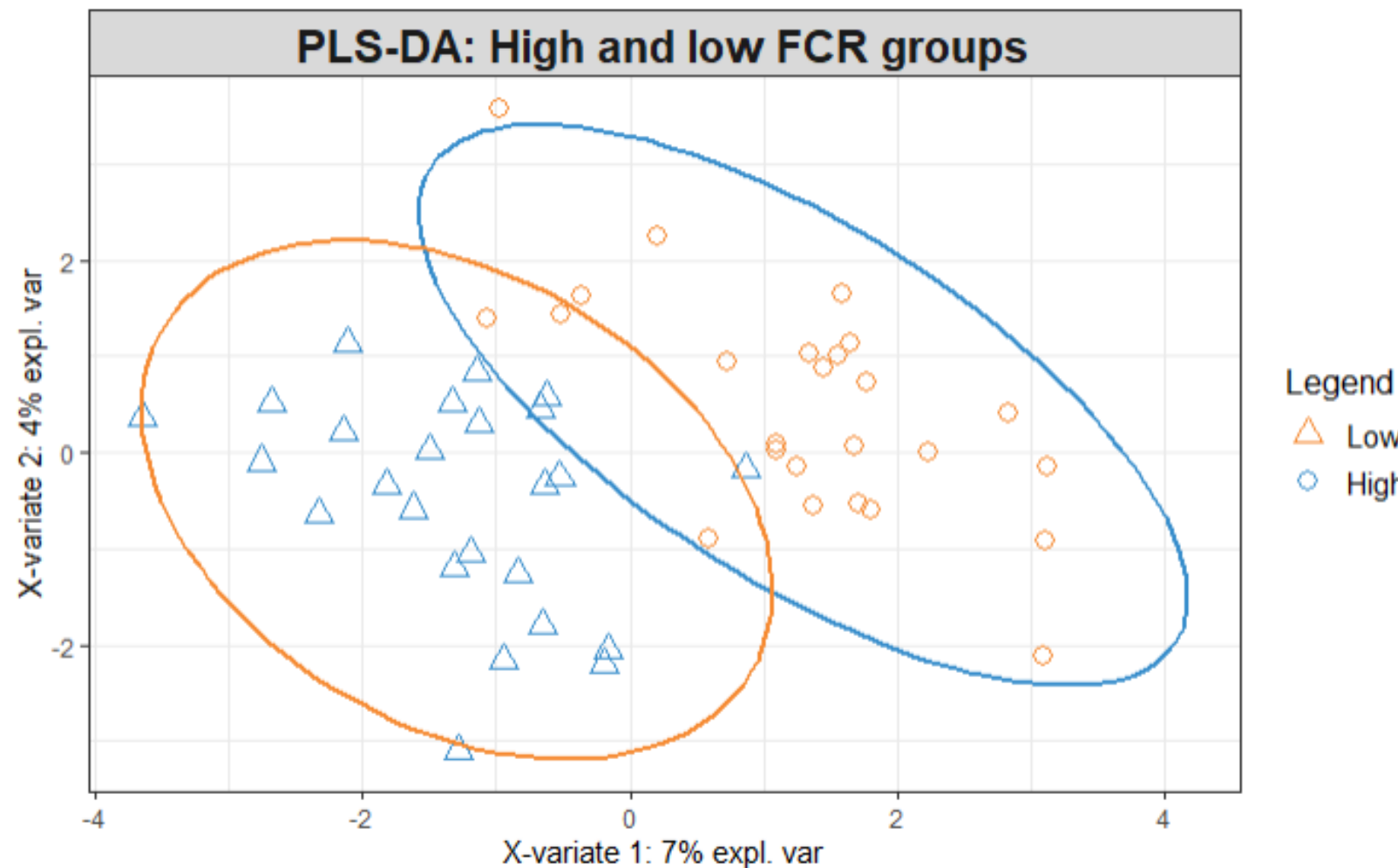
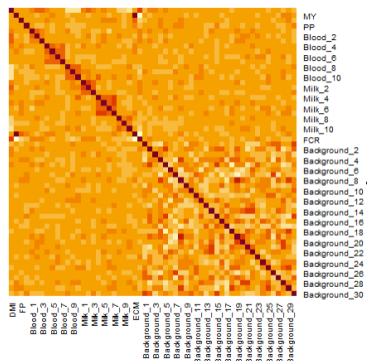
4. Simulation framework



5. Results and interpretation

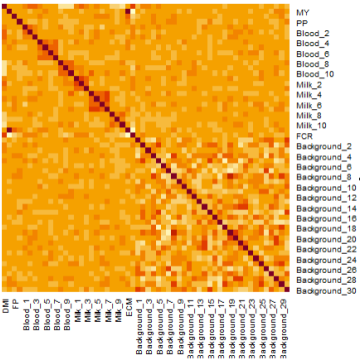
5. Results and interpretation

- PLS-DA analysis



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- PLS-DA analysis



	Biomarker	VIP
Milk_9	Milk_9	2.3465168
Blood_9	Blood_9	2.1873574
Blood_8	Blood_8	2.0715357
Blood_10	Blood_10	2.0586517
Background_23	Background_23	1.7705476
Milk_10	Milk_10	1.7026188
Background_1	Background_1	1.6037186
Milk_7	Milk_7	1.5744965
Milk_8	Milk_8	1.3353507
Background_29	Background_29	1.1870888

- *Essential*

- Gaynor, R.C., Gorjanc, G., & Hickey, J.M. 2021. AlphaSimR: an R package for breeding program simulations. G3. <https://doi.org/10.1093/g3journal/jkaa017>
- Grund, B. & Sabin, C. (2010). Analysis of biomarker data: logs, odds ratios, and receiver operating characteristic curves. Current Opinion in HIV and AIDS. <https://doi.org/10.1097/COH.0b013e32833ed742>

- *Further*

- Marina, H., Arranz, J.J., Suárez-Vega, A., Pelayo, R., Gutiérrez-Gil, B., Toral, P.G., Hervás, G., Frutos, P. & Fonseca, P.A.S. (2024). Assessment of milk metabolites as biomarkers for predicting feed efficiency in dairy sheep. Journal of Dairy Science. <https://doi.org/10.3168/jds.2023-23984>
- Foldager, L., Gaillard, C., Sørensen, M.T., Larsen, T., Matthews, E., O'Flaherty, R., Carter, F., Crowe, M.A., Grelet, C., Salavati, M., Hostens, M., Ingvarlsen, K.L., Krogh, M.A. & GplusE Consortium (2020). Predicting physiological imbalance in Holstein dairy cows by three different sets of milk biomarkers. Preventive Veterinary Medicine. <https://doi.org/10.1016/j.prevetmed.2020.105006>

hands
On



SCIENCE AND
EDUCATION **FOR**
SUSTAINABLE
LIFE